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Decision trees

Week 2: and explainable biological prediction

Dr Haithem Afli

Department of Computer Science

Haithem.afli@mtu.ie

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This lecture

- Act 1 Why trees? From Guess Who to biological samples
- Act 2 How does a tree choose? Entropy and information gain
- Act 3 From idea to algorithm: ID3, thresholds, and what scikit-learn actually does
- Act 4 Real biological data: a tree fitted to 398 diagnostic cases
- Act 5 Can we trust the tree? Depth, overfitting, pruning, validation
- Act 6 From model to application: deploy it, and interrogate it

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Where Week 2 sits

One method, taken all the way from intuition to a working application.

	Week	What it adds
Before	1	What machine learning is; why biological data is hard; foundation models, conceptually
This week	2	The first real model: grow one, read it, break it, prune it, deploy it
Next	3	Whether the number you just reported can be believed: leakage, baselines, cross-validation
Later	10	Many trees at once: bagging, random forests, feature importance

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Why trees?

Part 1 · From Guess Who to biological samples

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The biological problem

Wisconsin Diagnostic Breast Cancer: 569 cases, 30 features, 212 malignant and 357 benign.

- A fine-needle aspirate is taken from a breast mass and imaged
- The boundary of each cell nucleus in the image is traced
- Ten measurements are computed per nucleus: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension
- Each is summarised across the image three ways: its mean, its standard error, and its worst (largest) value
  - Ten measurements x three summaries = 30 features per case
- The label is the diagnosis: benign or malignant

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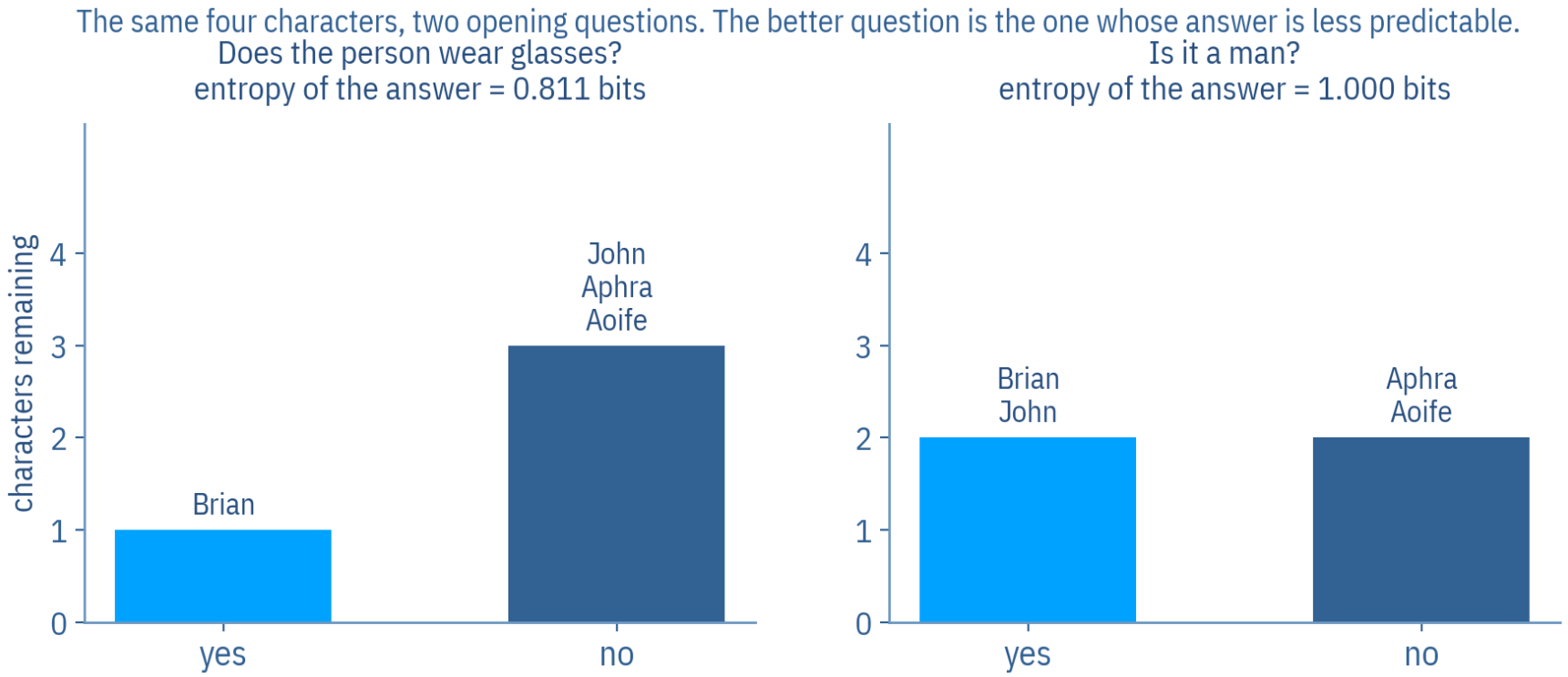
The features are already meaningful numbers.

Nobody needs a pretrained encoder to turn a nuclear radius into a number: a pathologist already did. That is why Week 2 is a classical machine-learning week, and it is the first case the module makes for choosing the simplest appropriate method.

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Anyone for a game of Guess Who?



Four characters, described by three binary features. Which question would you ask first: does the person wear glasses, or is it a man?

The entropies are computed from these four characters.

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The four characters

Man	Long hair	Glasses	Name
Yes	No	Yes	Brian
Yes	No	No	John
No	Yes	No	Aphra
No	No	No	Aoife

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So which question should we start with?

- On average, an answer to "is it a man?" gives you more information than an answer to "does the person wear glasses?"
- It needs fewer follow-up questions
- This is not because of the literal message of the answers, yes or no
- It is to do with how the answer splits the domain into different sized sets, and the likelihood of each possible answer

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Find the features whose answers are least predictable, and ask about those first.

The big idea, from original slide 10. Everything in Act 2 is one way of measuring "least predictable", and everything in Act 3 is the algorithm that uses it.

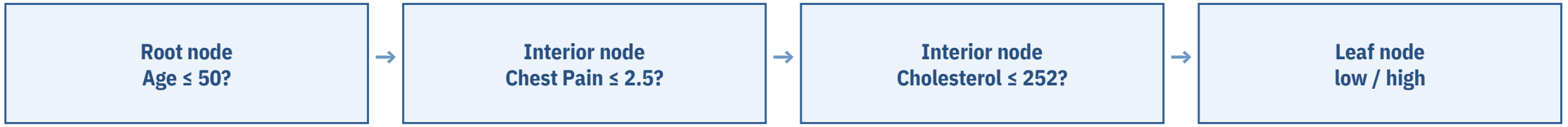
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Root, interior nodes, leaves

A decision tree produces a prediction by testing a series of descriptive feature values. The root is where the first test happens; interior nodes hold further tests; leaves specify the prediction.

The path a query follows



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One query travelling through the tree

Follow a path based on the descriptive feature values of a query to make a prediction.

- Age: 62
- Cholesterol: 311
- Chest Pain: 4
- Blood Pressure: 130
  - Age ≤ 50? No, so take the right branch
  - Chest Pain ≤ 2.5? No, so take the right branch
  - The leaf says: high risk

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But we can build many trees

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Both give similar predictions for query instances. Which would you choose?

Tree (a)

- Tests Age first, then Chest Pain
- Three tests deep on the longest path
- Uses three of the four features

Tree (b)

- Tests Blood Pressure first
- Five tests deep on the longest path
- Uses all four features, some of them twice

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Inductive bias

- Apply the same approach as in Guess Who: prefer simpler trees that use fewer tests
- This preference is an example of inductive bias: a set of assumptions that define the model selection criteria of a learning algorithm
- Restriction bias limits the set of models the algorithm can express
- Preference bias orders the models the algorithm can express
- Decision-tree learning uses a preference bias
- Inductive bias is necessary for generalisation

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How do we create simple trees?

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By testing the most informative features early.

Informative features split the dataset into pure sets with respect to the target. So all we need is a computational measure of the purity of a set. That measure is entropy.

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How does a tree choose?

Part 2 · Entropy, and information gain

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Shannon's entropy

$$H(S) = -\sum p(c) \cdot \log_2 p(c)$$

A weighted sum of the logs of the probabilities of each category in a set. High probability gives low entropy; low probability gives high entropy. Multiplying the log by −1 produces that mapping.

- **S** the set of labels at a node
- **p(c)** the proportion of the set carrying class c
- **$\log_2$** base 2, so entropy is measured in bits
- **H = 0** every element has the same label: the set is pure
- **H = 1** two classes, split exactly evenly: maximum uncertainty

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Pure and mixed nodes, on real cases

REAL DATA. Computed on the 398 training cases by data/scripts/06_build_week02.py.

A node containing	Benign	Malignant	Entropy	Reading
all 398 training cases	250	148	0.952 bits	close to maximally uncertain
the left branch of the root	241	19	0.377 bits	mostly benign, some doubt left
the right branch of the root	9	129	0.348 bits	mostly malignant
a pure leaf of the fitted tree	0	113	0.000 bits	no uncertainty at all

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Worked example: heart-disease risk

Thirteen patients. Every entropy on the next slides is computed from this table, so you can check any of them by hand.

ID	Age	Chest pain	Blood pressure	Cholesterol	Risk
1	53	3	135	160	High
2	52	2	140	216	Low
3	32	3	110	250	Low
4	52	2	150	215	Low
5	53	2	130	220	Low
6	60	1	120	291	High
7	49	2	140	238	Low
8	55	4	140	268	High
9	24	4	100	253	Low
10	39	2	160	147	Low
11	35	1	140	167	Low
12	49	2	142	207	Low
13	31	2	120	273	Low

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One split, worked in full

$$H(S) = -(3/13)\log_2(3/13) - (10/13)\log_2(10/13) = 0.779 \text{ bits}$$

That is the entropy before we ask anything: 3 high-risk and 10 low-risk patients in 13. Now split on Age at 52.5 and measure what is left.

remainder the weighted average entropy of the two partitions, 0.250 bits

information gain 0.779 − 0.250 = 0.530 bits

weighted each partition counts in proportion to how many patients land in it

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All four features, compared

Entropy before any split: 0.779 bits. Age separates these patients almost cleanly; blood pressure hardly separates them at all, so the ordering matches clinical intuition.

Feature	Best threshold	Entropy after	Information gain
Age	≤ 52.5	0.250 bits	0.530 bits
Cholesterol	≤ 260.5	0.573 bits	0.207 bits
Chest Pain	≤ 2.5	0.656 bits	0.123 bits
Blood Pressure	≤ 141.0	0.678 bits	0.101 bits

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Information gain

The entropy before partitioning, minus the average entropy of the partitions after it.

When building a tree you can work with the remainder directly and never compute the gain; the gain is the more intuitive quantity to talk about, because it answers "how many bits of uncertainty did that question remove?"

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Workshop: which feature belongs at the root?

Six patients, three binary descriptive features. Which feature gives the partitions with the lowest average entropy? Five minutes, in pairs.

ID	Obese	Smoker	Drinks alcohol	Cancer risk
1	true	false	true	low
2	true	true	true	high
3	true	false	true	low
4	false	true	true	high
5	false	true	false	low
6	false	true	true	high

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Workshop: the answer

Entropy before the split: 1.000 bit (3 high, 3 low). Smoker wins, with an information gain of 0.459 bits.

Split on	Level	Counts	Entropy	Weighted	Gain
Smoker	true	3 high, 1 low	0.811	0.541	0.459
	false	0 high, 2 low	0.000		
Drinks alcohol	true	3 high, 2 low	0.971	0.809	0.191
	false	0 high, 1 low	0.000		
Obese	true	1 high, 2 low	0.918	0.918	0.082
	false	2 high, 1 low	0.918		

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From idea to algorithm

Part 3 · ID3, thresholds, and what scikit-learn actually does

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The ID3 algorithm

- Iterative Dichotomiser 3
- Attempts to create the shallowest tree that is consistent with the data it is given
- Builds the tree recursively and depth-first, from the root down to the leaves
  - Choose the best descriptive feature to test, using entropy
  - Add a root node labelled with that feature
  - Partition the training data using the test
  - Grow a branch from the node for each partition
  - Repeat on each branch, with that partition in place of the full dataset and the selected feature excluded from further testing

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Stopping rules

The three situations in which standard ID3 stops.

- All of the instances in the dataset have the same classification: return a leaf labelled with that classification
- The set of features left to test is empty: return a leaf labelled with the majority class of the dataset
- The dataset is empty: return a leaf labelled with the majority class of the dataset at the parent node that made the recursive call

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Applying it to the heart-disease set

The point is the recursion, not the bookkeeping.

- Age wins the root, as the table showed
- The Age ≤ 52.5 partition is pure: every patient in it is low risk, so it becomes a leaf immediately
- The other partition is mixed, so the procedure recurses on it
- Within that partition the entropies are recomputed, on those patients only, and a new winning feature is chosen
- The recursion continues until one of the three stopping rules fires
- In the original deck this produced a tree testing Age, then Cholesterol, then Chest Pain

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Using the tree

- Age: 59 Cholesterol: 225 Chest Pain: 4 Blood Pressure: 142
- The tree returns a prediction for this query
- Note that there was no example in the training data with exactly this combination of feature values
- The model has generalised: it is applying a rule learned from other patients to one it has never seen
- That is the whole point, and it is also where every failure in this module begins

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Handling continuous features

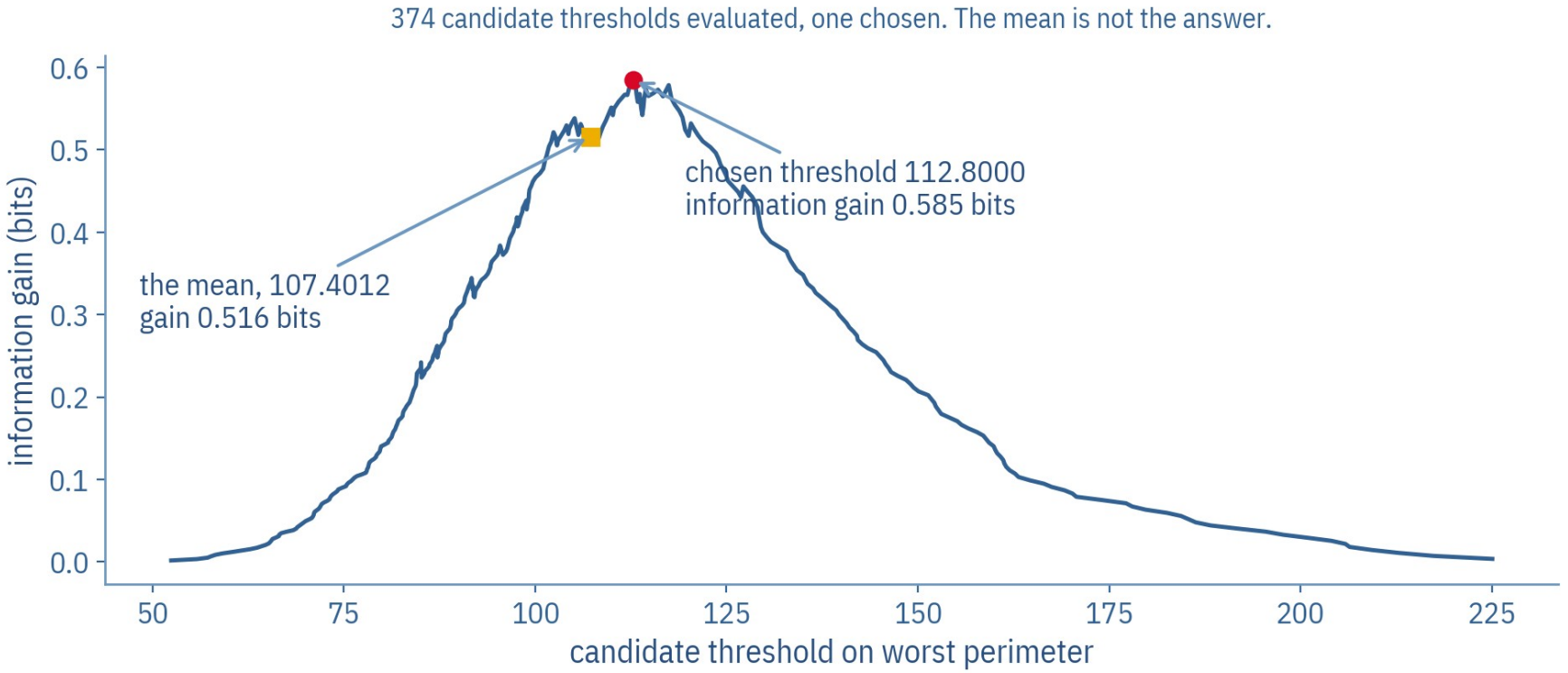
Answered with a measurement on the next slide.

- In all the examples so far, the threshold for a continuous feature was set to the mean value of that feature
- This is not always the best approach
- It is typical to calculate the average partition entropy using every potential threshold
- Then choose the threshold that gives the lowest average entropy in the resulting partitions
  - Candidate thresholds sit midway between adjacent observed values: 12.3, 13.7, 15.1, 16.8, 18.2 and so on

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Every candidate threshold, evaluated



374 candidate thresholds on worst perimeter. The algorithm takes 112.8, worth 0.585 bits. The mean, 107.4, would have been worth 0.516.

REAL DATA. Computed on the 398 WDBC training cases by data/scripts/06_build_week02.py.

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ID3 is not what scikit-learn runs

Both are decision-tree learners. They are not the same algorithm, and the lab uses the second one.

ID3, as taught above

- Chooses splits by information gain
- Historically associated with categorical features
- Splits a categorical feature into one branch per level, so a node can have many children
- Removes a feature from consideration once it has been used
- No built-in pruning in the standard formulation

sklearn DecisionTreeClassifier

- CART-style recursive binary splitting: every node has exactly two children
- criterion='gini' by default, criterion='entropy' available
- Handles continuous features by searching thresholds, as above
- A feature can be tested again further down the same path
- Supports pre-pruning (max_depth, min_samples_leaf) and cost-complexity post-pruning (ccp_alpha)

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Real biological data

Part 4 · A tree fitted to 398 diagnostic cases

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The experiment, stated before it is run

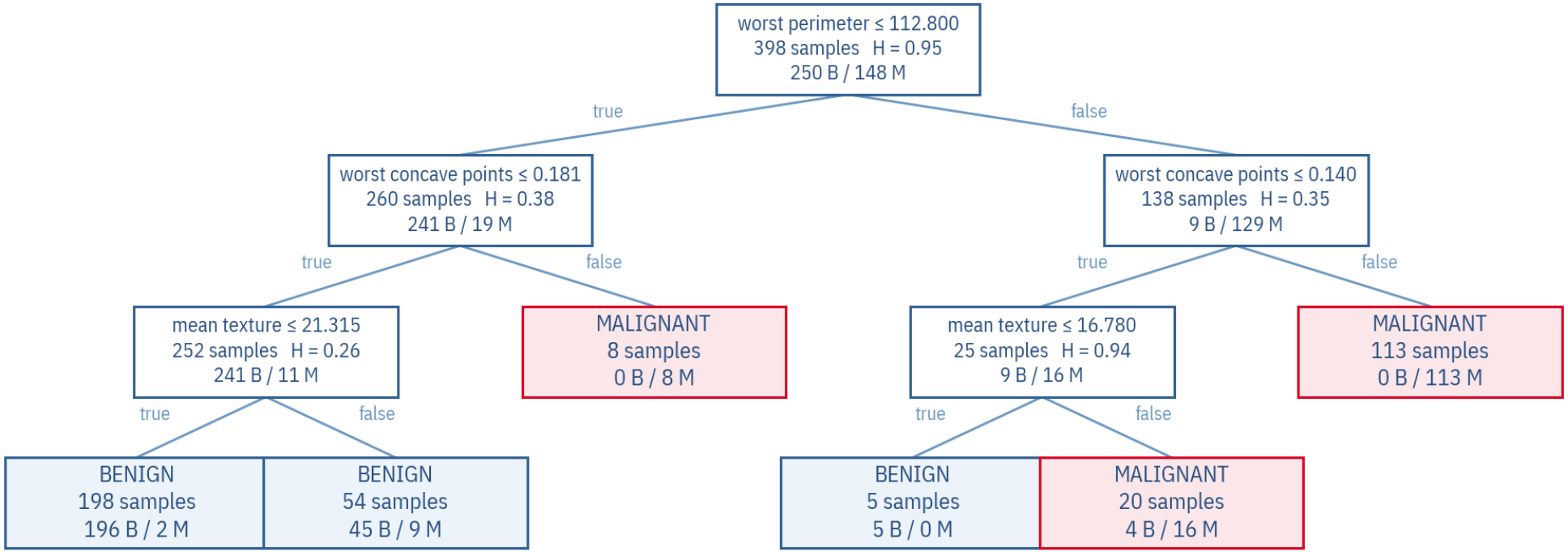
Write the protocol down before you see a number. Week 3 is about what happens when you do not.

Decision	What it is, and why it is fixed now
Dataset	Wisconsin Diagnostic Breast Cancer, 569 cases, 30 features
Source	sklearn.datasets.load_breast_cancer (UCI)
Classes	212 malignant, 357 benign
Majority baseline	62.7 per cent: predict benign for everything
Split	stratified 70 / 30, random_state=42
Training / held out	398 cases / 171 cases
Model	DecisionTreeClassifier(criterion='entropy', max_depth=3)
Provenance	REAL benchmark data. Not a clinical system

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The actual fitted tree



Each node: the split, the training cases reaching it, the entropy, and the benign / malignant counts. The fitted object, not a drawing of one.

REAL DATA. criterion='entropy', max_depth=3, random_state=42. Training accuracy 96.2 per cent, held-out accuracy 91.8 per cent.

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The root split, checked by hand

```
H(before)  =  0.952          H(after)  =  0.367
        information gain  =  0.585 bits
```

worst perimeter ≤ 112.8. The left partition holds 260 cases at 0.377 bits; the right holds 138 at 0.348 bits.

checked the same arithmetic as the heart-disease example, on 398 cases instead of 13

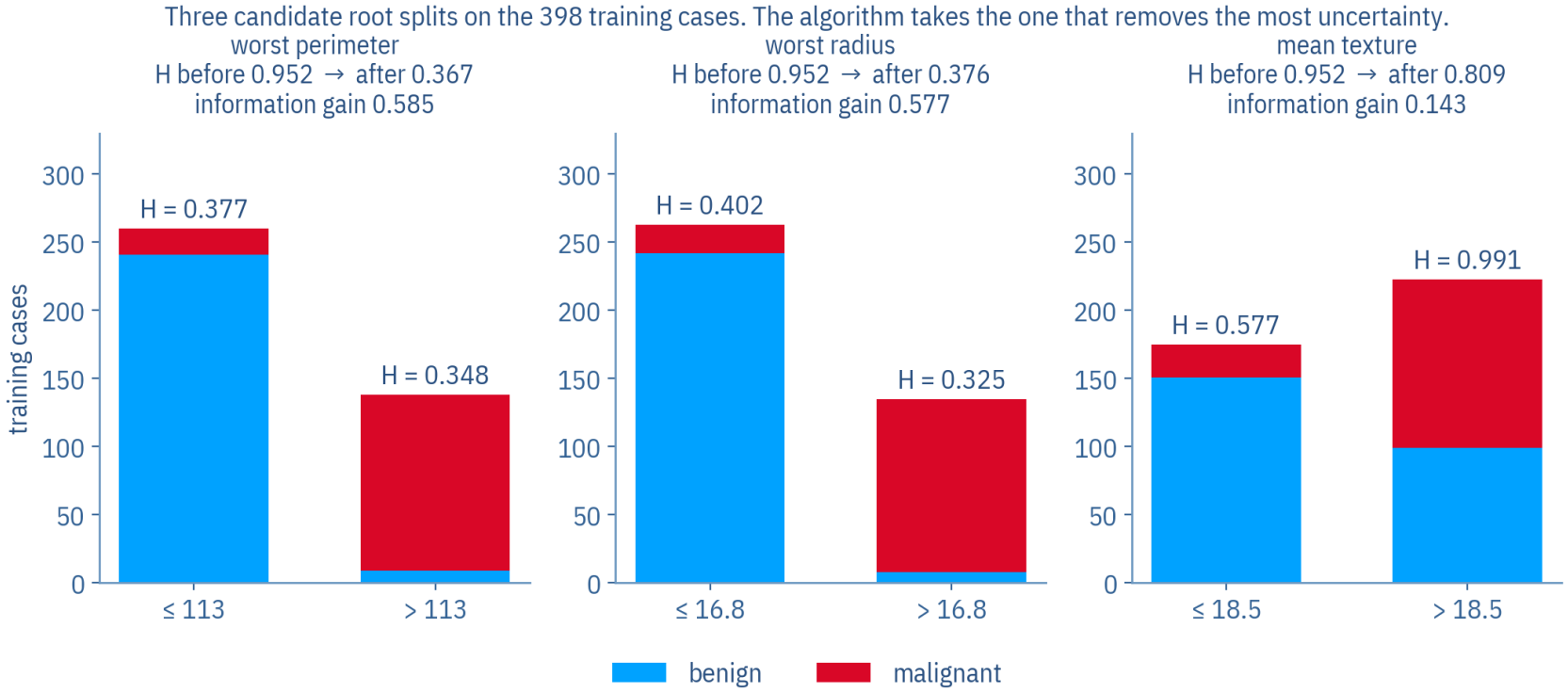
biologically perimeter of the largest nucleus in the image: larger nuclei are associated with malignancy

but association in this dataset, not a mechanism, and not a diagnostic threshold

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Three candidate splits at the root



The algorithm evaluated every feature at every threshold and took the first of these. Mean texture is a perfectly sensible biological measurement and a poor first question.

REAL DATA. Computed on the 398 training cases.

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One real held-out case

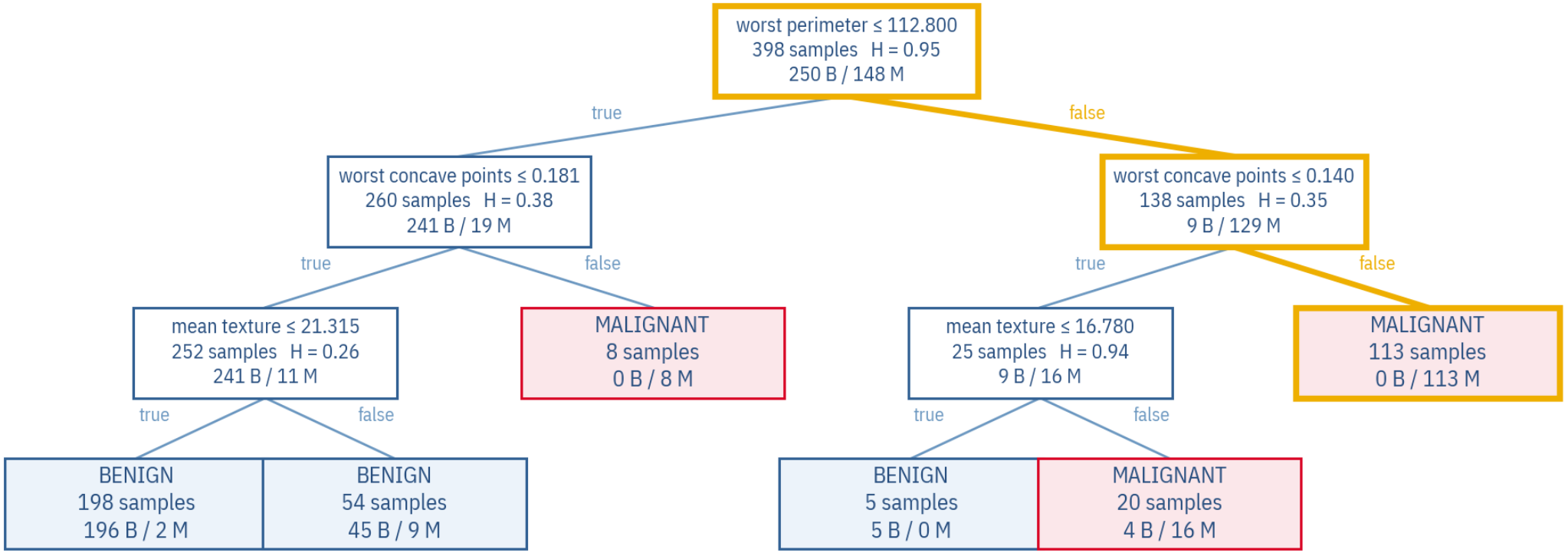
The same move as the heart-disease query, on a real biopsy the model has never seen.

- Case test-006 from the held-out set, truly malignant
- worst perimeter = 152.200, threshold 112.800 → false
- worst concave points = 0.163, threshold 0.140 → false
- Leaf: MALIGNANT, built from 113 training cases, 0 benign and 113 malignant

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The path, on the tree



Two tests, and the case lands in a leaf built from 113 training cases, all of them malignant.

REAL DATA. The same fitted tree, the same held-out case.

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Can we trust the tree?

Part 5 · Depth, overfitting, pruning, validation

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Overfitting

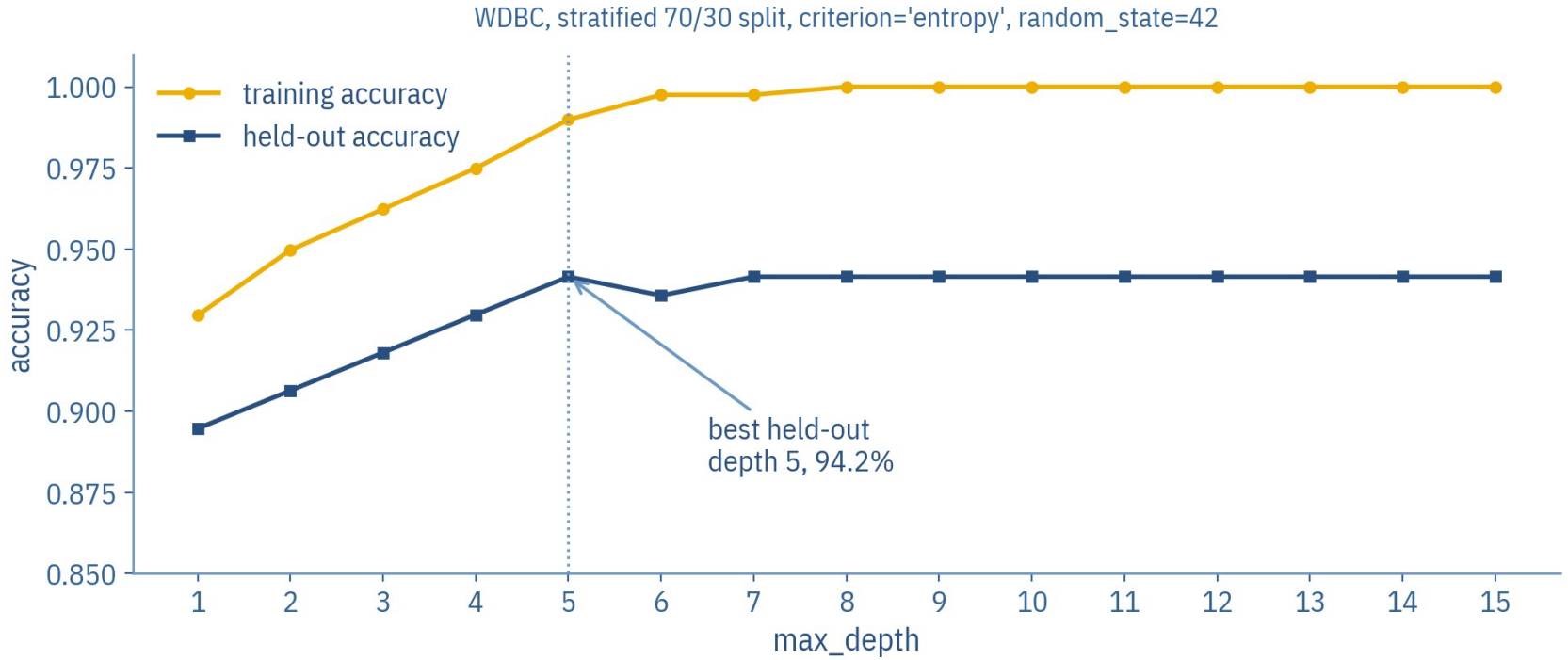
And the measurement that shows it happening.

- In a decision tree, overfitting involves splitting the data on an irrelevant feature
- The likelihood increases as a tree gets deeper, because the resulting classifications are based on smaller and smaller subsets as the dataset is partitioned after each test in the path
  - The unrestricted tree on this data reaches depth 8 with 16 leaves
  - It classifies 100.0 per cent of the training cases correctly
  - On the held-out cases it gets 94.2 per cent

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Depth against performance, measured



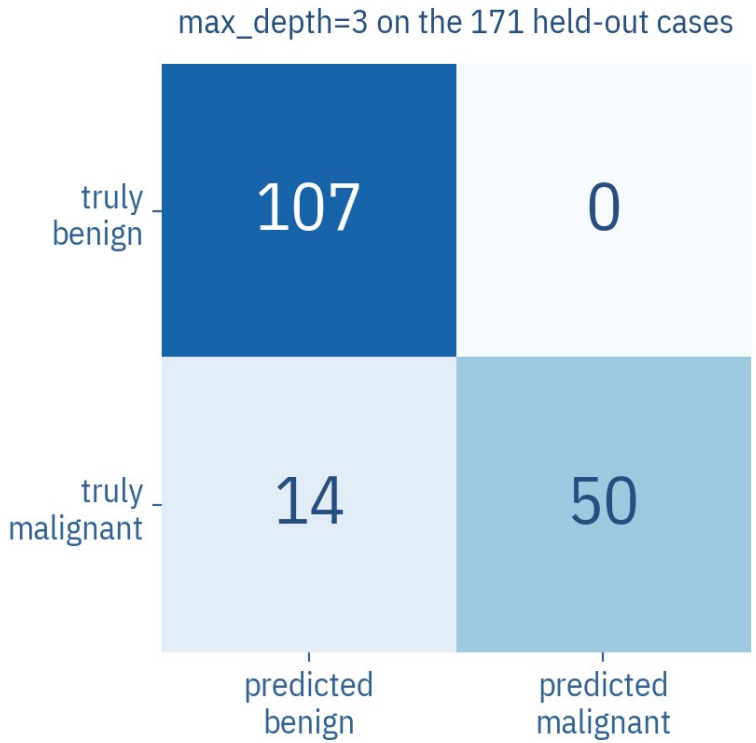
Training accuracy climbs to 100 per cent. Held-out accuracy reaches 94.2 per cent at depth 5 and never improves on it again: every extra split after that buys training accuracy and nothing else. The gap between the curves is the overfitting.

REAL DATA. Every point is a tree fitted on the same 398 cases and scored on the same 171.

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Where the errors are



107 true negatives
correctly called benign

0 false positives
benign, called malignant: an unnecessary intervention

14 false negatives
malignant, called benign: a missed cancer

50 true positives
correctly called malignant

Accuracy counts all four equally. Nothing else does.

Accuracy 91.8 per cent hides the shape of the errors. This tree makes 14 false negatives and 0 false positives, and those two numbers do not cost the same thing.

REAL DATA. max_depth=3 on the 171 held-out cases.

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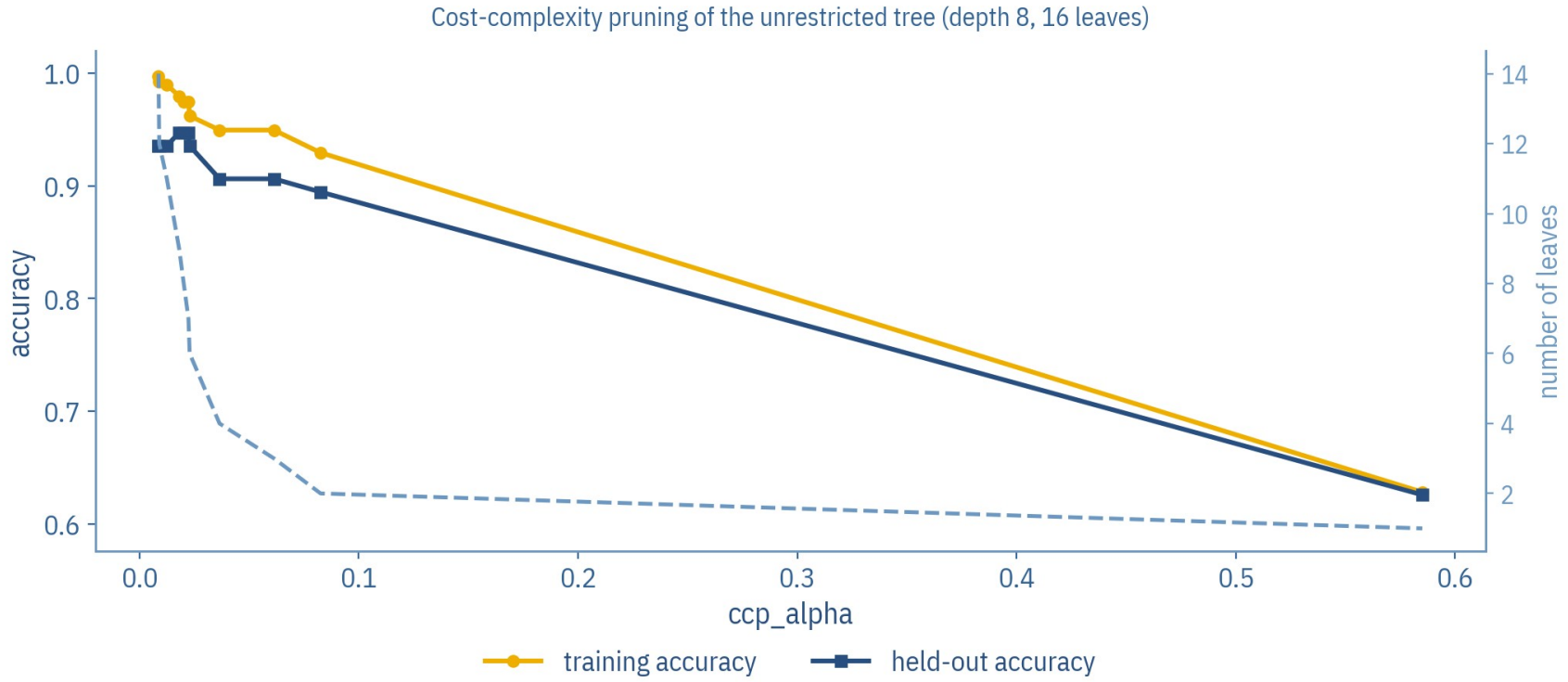
Pruning

- Pre-pruning: stop the recursive partitioning early, typically using rules about the number of instances allowed at a leaf
  - max_depth, min_samples_leaf, min_samples_split
- Post-pruning: let the algorithm grow the tree as much as it likes, then prune away the branches that cause overfitting
  - A common approach: evaluate on a validation set as the tree grows, and prune back when training and validation performance diverge
  - scikit-learn's version is cost-complexity pruning, ccp_alpha
- Advantages: smaller trees are easier to interpret, and generalisation improves when there is noise in the training data

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Pruning, measured on the same split



Raising ccp_alpha removes leaves. Training accuracy falls steadily; held-out accuracy rises first, peaking at 94.7 per cent with 9 leaves, then collapses when the tree is cut too far.

REAL DATA. Cost-complexity pruning of the unrestricted tree.

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We just chose a hyperparameter by looking at the test set. That number is no longer an honest estimate.

Every time you look at the final test set and change something, you fit a little of it. Split three ways: train, validate, and a final test set you touch exactly once. The lab does it properly.

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Decision trees: advantages and disadvantages

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Advantages

- Interpretable
- Handle both categorical and continuous descriptive features
- Can model interactions between descriptive features, though pre-pruning diminishes this
- Relatively robust to the curse of dimensionality
- Relatively robust to noise, if pruning is used

Disadvantages

- Trees become large when dealing with continuous features
- Very expressive and sensitive to the dataset, so they can overfit when there are many features
- Eager learner, so they are exposed to concept drift

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A model can be interpretable and still be wrong.

You can read every rule this tree uses. That tells you what it does. It tells you nothing about whether what it does is correct, whether the data it learned from represents the patients you care about, or what happens when the measurements come from a different laboratory.

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From model to application

Part 6 · Deploy it, then interrogate it

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What deployment actually means

The model does not change. What changes is that a person who did not fit it is now acting on its output. On the Week 2 page you can enter measurements or load one of four real held-out cases, and out-of-range values are flagged against the actual dataset range.

The notebook

Fit on 398 cases → **Score on 171 held out** → **Read the tree**

The application

A new case arrives → **The same fitted tree evaluates it** → **Prediction and decision path** → **Someone decides what to do**

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Try this interactively

Decision Tree Builder, Entropy Visualiser and the Deployment demonstration

Change the depth and the leaf size and watch the tree and the held-out accuracy change. Drag the split point and watch the entropy and information gain update. Then put a case through the tree and read its decision path.

`weeks/week02.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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Decision path is not an explanation

The same wording appears on the website and in the lab.

- Call it the decision path
- Do not call it the reasoning, or why the model thinks this
- The path shows the sequence of learned splits the tree used for this input
- It does not establish causality
- It does not provide a clinical explanation
- It is a faithful account of what the model did, and nothing more

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Should we trust this result?

Producing a prediction is not the end of machine learning. It is the point at which these questions start.

- Was the model validated, and on data it never saw during fitting?
- Was the final test set kept separate, and looked at once?
- Is the dataset representative of the patients this would be used on?
- Has any external validation been done, on another institution's data?
- Could dataset shift change the result: different scanner, different laboratory, different population?
- Are the features appropriate, and are they available at the moment the decision must be made?
- What are the consequences of a false positive, and of a false negative? They are not the same.
- Is this model clinically validated? No.

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Training a model is not the end of machine learning.

You have to inspect it, validate it, understand its behaviour, and know the limits of its predictions. That is what the rest of this module is about, and it is what both projects are marked on.

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Exit ticket, and the laboratory

Diagnostic Model Auditor: decision trees on real biological data. Two hours.

- Write down, in one sentence each:
  - Why did the tree ask about worst perimeter first?
  - What would make you doubt 91.8 per cent held-out accuracy?
  - What is the difference between the decision path and an explanation?
- In the laboratory you become the auditor: grow a tree, break it, prune it, trace a case, deploy it, and write the model card that says what it must not be used for

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Summary

- A decision tree makes predictions from sequences of tests on the descriptive feature values of a query
- Entropy measures the impurity of a set; information gain measures how much a test reduces it
- ID3 is the standard conceptual algorithm; scikit-learn's DecisionTreeClassifier is CART-style and is not the same thing
- Continuous features are handled by searching candidate thresholds, not by taking the mean
- Deeper is not better: on this data the held-out curve reaches its best at depth 5 and never improves on it, while training accuracy runs to 100 per cent
- Pruning trades training accuracy for generalisation and readability
- Interpretability tells you what a model did, not whether it is right

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References and further reading

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Questions

Next week: the complete workflow, and why the number you just reported may not survive contact with an honest protocol.

Dr Haithem Afli · haithem.afli@mtu.ie

Succeeding Together

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